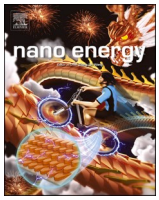
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Full paper



A visual-tactile synchronized stimulation ring system for sensory rehabilitation integrating triboelectric sensing and pneumatic feedback

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ABSTRACT

Stroke stands as a leading cause of disability, often resulting in sensory and motor impairments, particularly in upper limbs. Sensory rehabilitation is vital for functional recovery but is hindered by its reliance on healthcare professionals and the prioritization of motor recovery over sensory restoration. This study introduces a Triboelectric Sensor and Pressure Feedback Ring (TSPF-Ring) as a solution to bridge this gap by amalgamating triboelectric sensing and pneumatic feedback into a wearable device. The sensitive tactile sensor is capable of recognizing multi-dimensional changes such as pressure, proximity and texture, converting these into visual stimuli with a classification accuracy exceeding 99 %. Meanwhile, the pneumatic actuator provides adjustable tactile feedback within a 0–12 N range. A visual-tactile synchronized rehabilitation training utilizing the TSPF-Ring system was implemented to effectively enhance activation in the patient's sensorimotor cortex, as validated by electroencephalogram experiments. The results indicate that the TSPF-Ring holds promise in improving hand sensory function in stroke patients by promoting neural remodeling through synchronized visual-tactile stimulation, offering a novel wearable device solution for sensory rehabilitation.

1. Introduction

Stroke, an acute cerebrovascular condition, has emerged as one of the leading causes of disability worldwide[1,2]. Typically, stroke results in the necrosis of certain brain tissues, leading to a range of functional impairments, including cognitive, motor, and sensory deficits, with upper limb dysfunction being particularly severe[3,4]. Coordinated hand sensory and motor functions are crucial for normal upper limb movements; however, studies have shown that 50–80 % of stroke patients experience sensory impairments[5], with light tactile being most commonly affected. Specifically, these patients often struggle to perceive subtle tactile differences and are unable to accurately distinguish between object properties such as shape, size, and texture, making daily tasks more challenging. Therefore, the incorporation of sensory rehabilitation training in upper limb rehabilitation is critical for the effective recovery of motor function in these patients [6].

Repetitive tactile stimulation plays a crucial role in the restoration of sensory function in stroke patients[3,7,8]. However, current sensory rehabilitation methods for the upper limbs of stroke patients often require continuous involvement and supervision from healthcare professionals. For instance, rehabilitation therapists typically perform manual interventions such as mechanical pressing of fingers[9], acupuncture[10], and multi-sensory stimulation[3], which deliver non-quantified tactile stimuli. While these approaches demonstrate the

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positive impact of sensory rehabilitation on the recovery process, they are limited in terms of efficiency, sustainability, and consistency.

Wearable devices, due to their portability and ease of use, offer a promising platform for providing convenient sensory rehabilitation training for stroke patients[11–14]. However, most existing wearable rehabilitation solutions primarily focus on restoring hand motor function[15], such as assisted hand stretching exercises[16–19], with limited attention to the rehabilitation of sensory impairments. Additionally, studies have shown that interactions between specific sensory modalities, such as vision, and the sensorimotor system can induce neuroplasticity in the primary somatosensory cortex[20,21]. Such multi-dimensional sensory stimulation could more effectively enhance

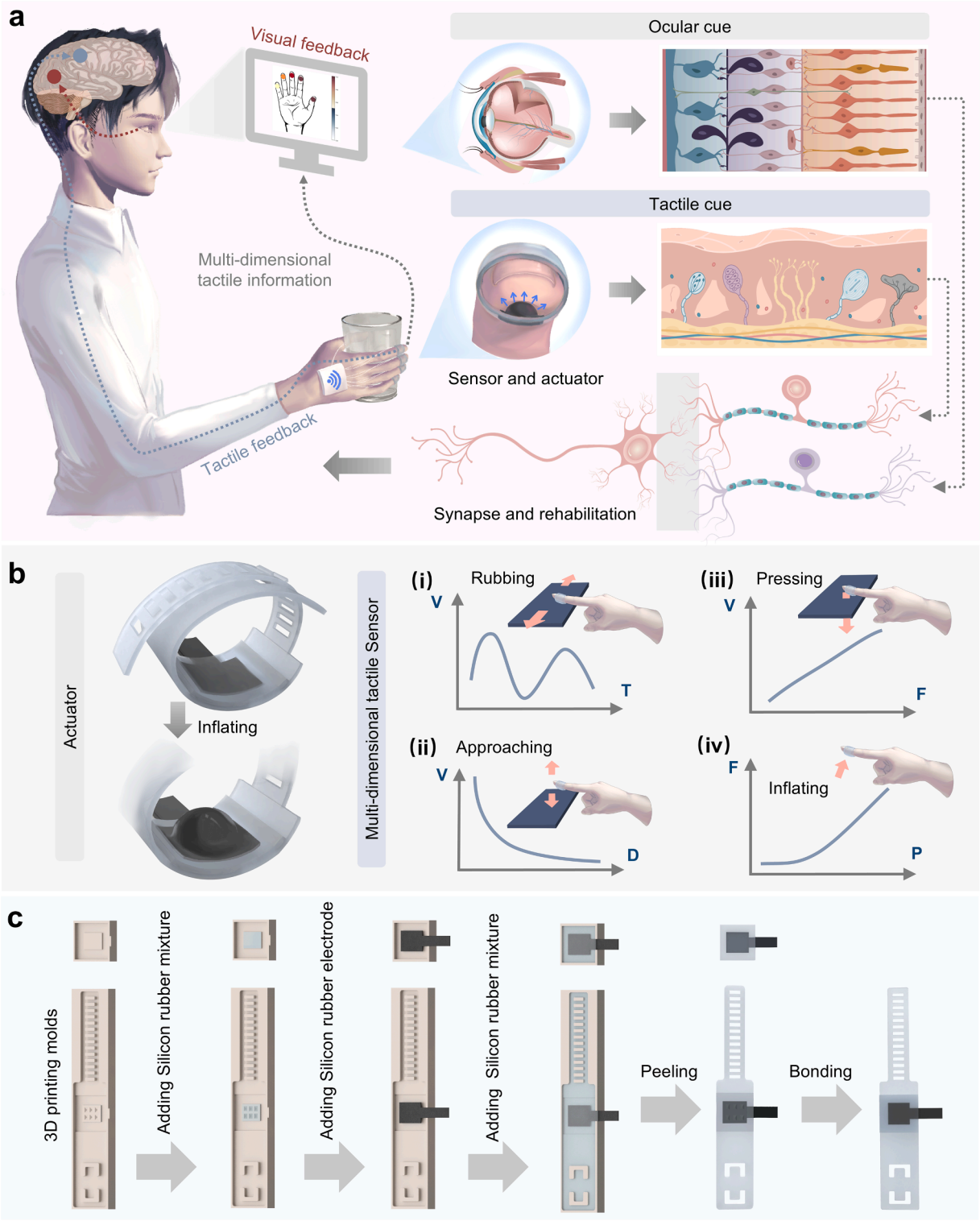


Fig. 1. System functionality and structure of the TSPF-Ring. (a) The TSPF-Ring integrates tactile sensing and pneumatic feedback mechanisms to promote neural pathway remodeling and functional recovery through visual-tactile stimulation. (b) The TSPF-Ring features actuation and multi-dimensional tactile information sensing capabilities. (c) The overall fabrication process of the TSPF-Ring.

stroke rehabilitation outcomes. Visual stimulation can be achieved by synchronously visualizing tactile information captured by sensors. Triboelectric-based sensing mechanisms, in particular, have attracted significant interest due to their heightened sensitivity and capacity to capture intricate tactile details[22–26]. However, in most wearable devices, sensors and actuators are typically designed as separate components[27–30]. As a result of integration limitations, the points of sensing and feedback action are often spatially separated. Therefore, integrating such sensing mechanisms with tactile feedback into a compact wearable device to create multiple forms of sensory stimulation for limb sensory rehabilitation in stroke patients remains a largely unexplored yet practical field with significant potential.

In this study, we propose a Triboelectric Sensor and Pressure Feedback Ring (TSPF-Ring), which integrates triboelectric-based sensing and pneumatic feedback stimulation into a single device. Through the integration of visual feedback derived from simultaneous sensory data with a pneumatic tactile feedback mechanism, this device promotes neural pathway remodeling and functional recovery via synchronized visual-tactile stimulation, offering a novel approach for sensory rehabilitation in stroke patients. The ring incorporates a triboelectric-based sensor, capable of highly sensitive detection of multi-dimensional tactile changes, such as pressure, proximity, and slip, with a signal recognition accuracy exceeding 99 % enabled by deep learning (DL), effectively compensating for light touch deficits in patients. The tactile feedback is generated via pneumatic actuation, inflating an air chamber to produce realistic and controllable tactile stimuli with an adjustable feedback pressure range of 0–12 N, ensuring quick response times and high stability. The TSPF-Ring enables closed-loop rehabilitation training for patients with sensory impairments. On one hand, the triboelectric sensor captures tactile information from the hand, such as pressure variations, and synchronously converts it into corresponding visual signals, like alterations in color, to activate the patient's central nervous system. On the other hand, pneumatic feedback simultaneously stimulates the peripheral nerves in the patient's fingers. Electroencephalogram (EEG) results demonstrate that this synchronized visual-tactile training significantly enhances activation in the brain's sensorimotor cortex, offering the potential for more effective hand sensory function recovery in stroke patients, thus improving their quality of life and rehabilitation outcomes.

2. Results and discussion

2.1. Design and architecture of TSPF-Ring

As shown in Fig. 1a, the TSPF-Ring integrates tactile sensing with a pneumatic feedback mechanism. The triboelectric-based high-sensitivity sensor effectively captures multi-dimensional tactile information from the fingers, including pressure, proximity, and texture, all within a flexible design. This tactile information could be visualized through wireless transmission, enabling patients to intuitively observe the distribution and changes in their hand's tactile sensations in real-time. This tactile discrimination training facilitates the reorganization of the sensorimotor cortex, aiding in the recovery of sensory function for patients[31].

Additionally, the ring incorporates an internal cavity structure that inflates through pneumatic pressure, providing pressure feedback to the fingertip and stimulating peripheral tactile nerves. By combining the visual feedback based on sensing data with direct tactile stimulation from the pneumatic mechanism, the device delivers synchronized visual-tactile stimulation. This closed-loop rehabilitation method effectively promotes neural pathway remodeling and functional recovery in patients with sensory impairments, enhancing their ability to perceive and process tactile stimulation.

Fig. 1b(i) illustrates the structural features of the TSPF-Ring before and after inflation. Fig. 1b(ii) demonstrates the ring's capability to capture multi-dimensional sensing signals, like slip, proximity, and pressure. These features enable a variety of intelligent recognition functions, which could potentially extend the application of tactile information in fields such as human-machine interaction and smart rehabilitation. The manufacturing process of the ring is shown in Fig. 1c, utilizing flexible silicone materials and drawing inspiration from watch straps to create a customizable and wearable form. Detailed fabrication methods can be found in the experimental section.

2.2. Working mechanism and characterization of TSPF-Ring

The detailed structure of the TSPF-Ring is shown in Fig. 2a, where its primary element is a dual-electrode triboelectric nanogenerator (TENG). Flexible silicone electrodes are fabricated as the upper and lower electrode layers of the sensor, with a silicone layer on the internal surface of the electrodes serving as the triboelectric layer. Details regarding the fabrication of these flexible silicone electrodes can be found in the experimental section and Supplementary Figure S1. The entire device is encapsulated with an additional layer of silicone for sealing, and the outer surface is treated with 120-grit sandpaper to create microstructures, reducing friction during sliding and improving both the stability of mechanical motion and the sensor's sensitivity[32]. Supplementary Figure S2 employs scanning electron microscope (SEM) images to show the difference between micro-structured and smooth surfaces of the materials.

An internal gap within the TSPF-Ring forms a cavity that serves as the foundation for pneumatic pressure feedback[33]. This structure ingeniously integrates triboelectric-based sensing and pneumatic feedback mechanisms into a single device, making it possible to develop an all-in-one wearable multi-sensory stimulation device. And it further guarantees stability and reliability over repeated usage. Subsequently, a linear motor was utilized to construct the testing platform for assessing the electrical capabilities of the sensor (Fig. 2b). Supplementary Figure S3 displays the output performance of the sensor under different configurations, including variations in material, area, surface structure, and applied pressure.

As illustrated in Fig. 2c, the proximity sensing principle of the TSPF-Ring operates on the triboelectric effect. Most surfaces carry a small amount of charge due to friction and the presence of air. When an object approaches the TSPF-Ring, the charges on its surface (e.g., a certain amount of positive charge) alter the electric field distribution between the upper and lower electrodes of the sensor, leading to a gradually increasing potential difference. This potential difference causes the transfer of charges in the output circuit, generating an electrical signal. To gain deeper insights into the alterations in potential throughout this process, electrostatic simulations using COMSOL were performed, which is consistent with theoretical analysis (Supplementary Figure S4).

The proximity of objects to the sensor was further characterized using a linear motor, facilitating the determination of the correlation between sensor output signals and the proximity of approaching objects, as illustrated in Fig. 2d. The sensor's range extends to 120 mm, with high sensitivity within the 0–20 mm span. In Fig. 2e, the sensor's electrical signal output at proximity frequencies ranging from 1 to 5 Hz is displayed, showcasing consistent signal amplitudes across varying frequencies. Figure S5 illustrates the stable signal output of the sensor over 10,000 of cycle tests, confirming its long-term durability. Moreover, materials of distinct compositions yield unique signal profiles when in close proximity to the sensor (Supplementary Figure S6), underscoring the potential utility of the TSPF-Ring in non-contact material recognition applications.

Fig. 2f illustrates the pressure sensing mechanism of the TSPF-Ring. The dual-electrode TENG employs identical friction materials for both the upper and lower triboelectric layers, effectively minimizing electrical signal fluctuations due to alterations in the internal electrode distance. This approach mitigates interference arising from variations in the air cavity volume on the sensing signal. As shown in Fig. 2g, a mere 0.08 V voltage signal shift is observed after pressurizing the air chamber

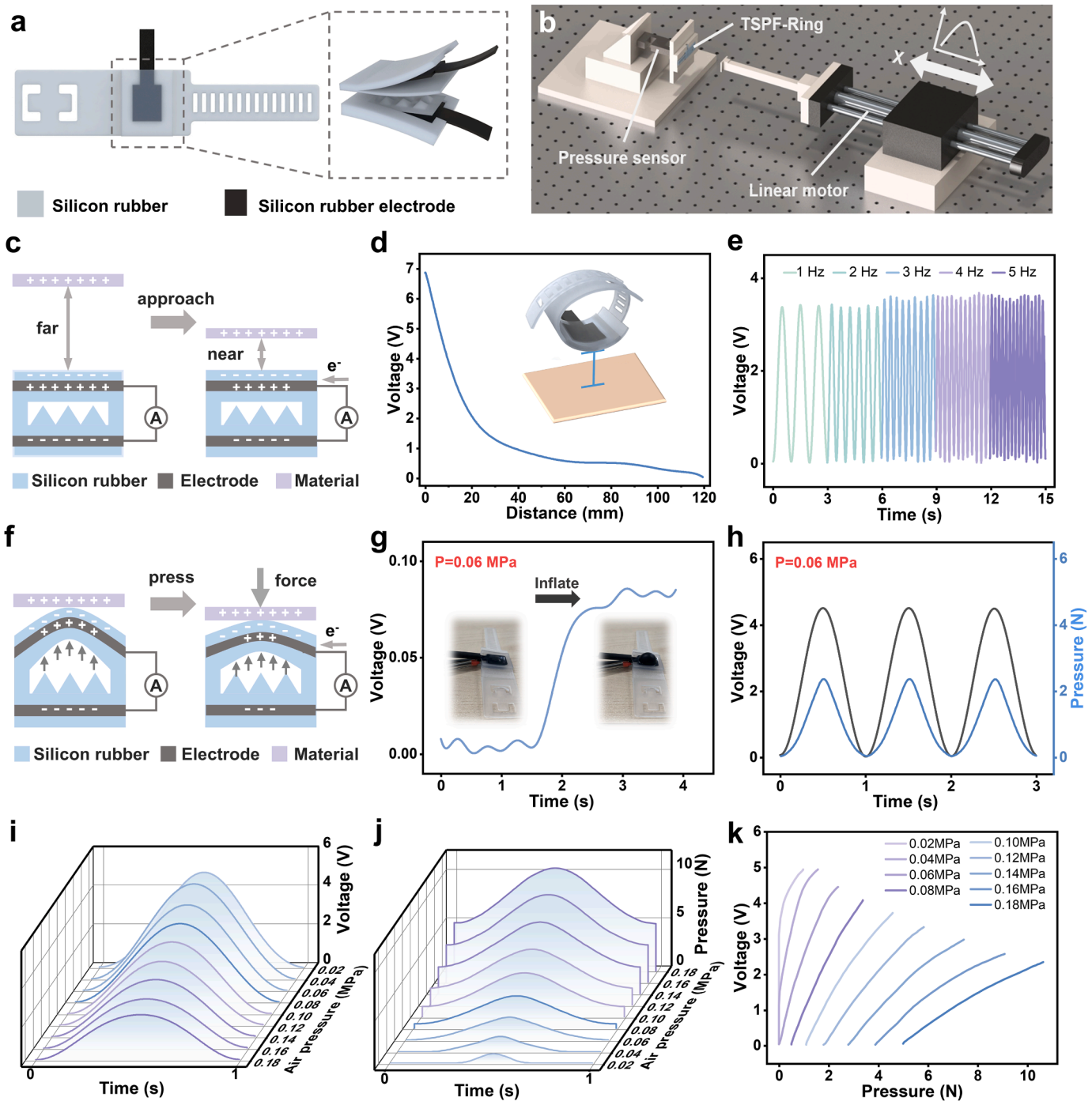


Fig. 2. Sensing principle and characterization of the TSPF-Ring. (a) Detailed structure of the TSPF-Ring. (b) Schematic of the test platform. (c) Schematic illustration of the proximity sensing principle. (d) Relationship curve between sensor output and proximity distance. (e) Electrical signal output of the sensor at different proximity frequencies. (f) Schematic illustration of the pressure sensing principle. (g) Minimal impact of air chamber inflation on the sensor's electrical signal output. (h) Synchronized changes between applied pressure and output voltage signal. (i) Voltage signal changes of the sensor under different air pressures. (j) External pressure changes under different air pressures. (k) Relationship curves between output voltage and corresponding external pressure at different air pressures.

to 0.06 MPa. When the air chamber inflates, the TSPF-Ring forms a large contact surface with the object, enhancing the pressure contact area. As the applied pressure increases, the object's surface and the air chamber's convex surface come into tighter contact. Similar to the proximity sensing principle, the charges on the object's surface cause an increasing potential difference between the TSPF-Ring's electrodes, which drives charge flow in the external circuit to generate an electrical signal. Supplementary Figure S7 shows COMSOL simulations of the changes in potential distribution during this process.

Next, the pressure sensing characteristics of the TSPF-Ring were analyzed. As shown in Fig. 2h, a distinct relationship between applied pressure and the voltage signal was evident at 0.06 MPa air pressure, affirming the viability of this design for pressure sensing applications. Subsequently, the sensor's output voltage signals were recorded across a spectrum of air pressures ranging from 0.02 to 0.18 MPa (Fig. 2i) alongside the associated pressure variations (Fig. 2j). More detailed data curves can be found in Supplementary Figure S8. Furthermore, Fig. 2k illustrates the relationship between output voltage and applied pressure.

It was observed that for air pressures equal to or exceeding 0.06 MPa, the output voltage exhibited a nearly linear relationship with the applied pressure, albeit with a diminishing slope as the air pressure rose. This suggests that increasing the air pressure can expand the pressure detection range of the TSPF-Ring, but with a corresponding decrease in sensitivity. This is because higher air pressure increases the range of contact area changes, while making the deformation of the contact surface more difficult. Through adjustments in air pressure, the range and sensitivity of pressure sensing can be tailored to individual preferences.

Figure S9 characterizes the pneumatic feedback properties of the TSPF-Ring, showing the adjustability of the feedback pressure within a span of 0–12 N with escalating inflation pressure. Supplementary Figure S10 presents a dual-section TSPF-Ring structure with an added fiber constraint layer, which enables bending actuation when the air cavity is inflated, assisting finger movements in patients with motor impairments. Video S1 provides a visual demonstration of the bending actuation capability of the device.

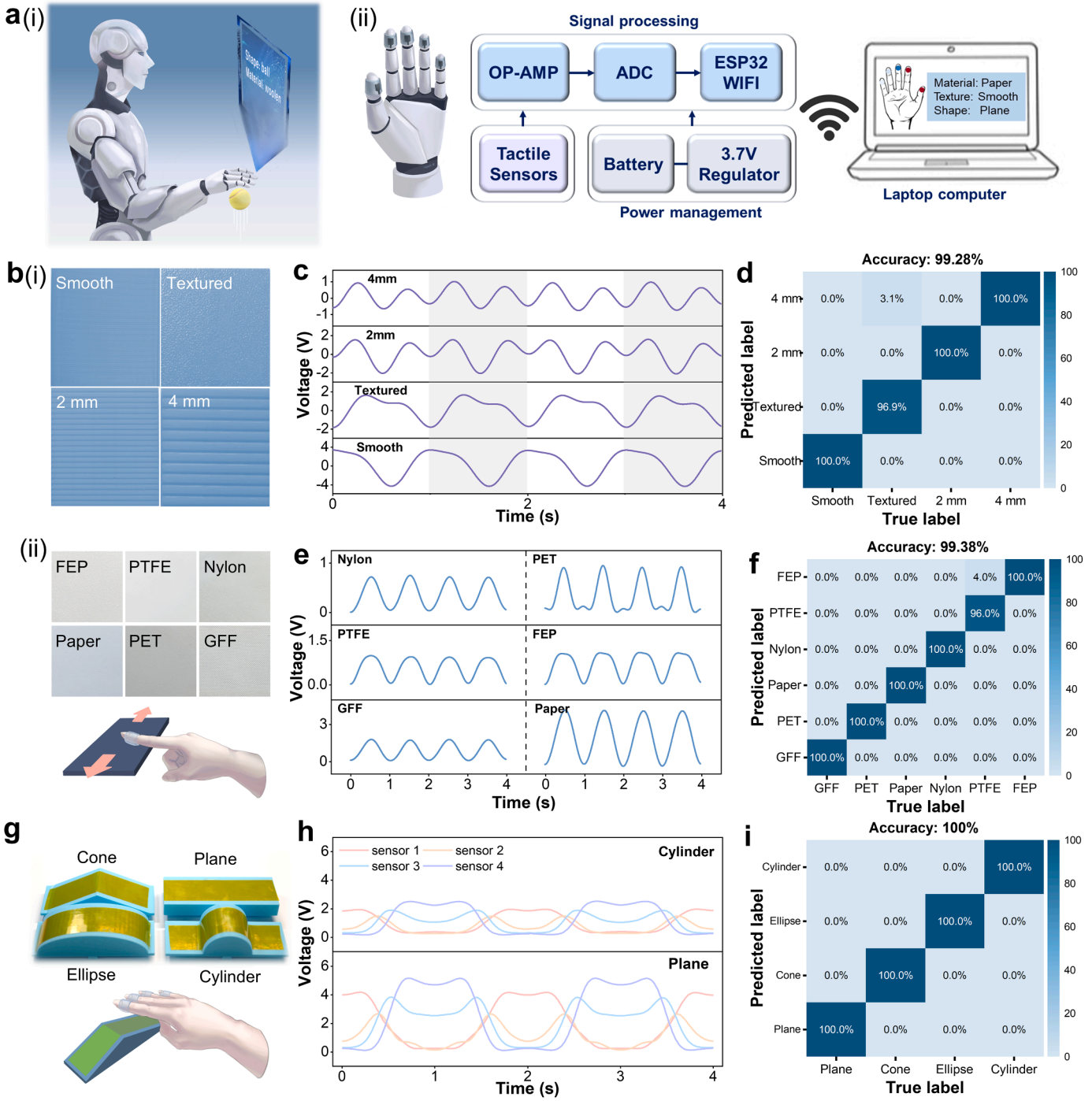


Fig. 3. Intelligent recognition system based on the TSPF-Ring. (a) (i) Intelligent recognition of object material, texture, and shape characteristics. (ii) Schematic of the system structure. (b) Contact-based recognition for (i) texture and (ii) material. (c) Sensing signals collected from four different surface textures. (d) Confusion matrix for texture recognition. (e) Sensing signals collected from six different materials. (f) Confusion matrix for material recognition. (g) Non-contact recognition of four approaching shapes using a sensor array. (h) Signals generated by the sensor array as a cylinder and plane surface passes. (i) Confusion matrix for the recognition of four approaching surface shapes.

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2.3. Deep learning-enabled intelligent recognition system

Patients with sensory impairments often experience tactile deficits. Research has demonstrated that combining tactile discrimination feedback with visual feedback training can effectively promote the recovery of sensory abilities in such patients[20,31,34]. In response to this need, an intelligent recognition system was developed based on the TSPF-Ring to assist patients in distinguishing corresponding tactile features. Through the application of deep learning algorithms, this system interprets signals for diverse classification tasks, encompassing texture, material, and object shape (Fig. 3a(i)). The primary structure of the system is illustrated in Fig. 3a(ii). The multi-channel TSPF-Ring generates diverse sensing signals during different tasks, which are collected and transmitted by a signal processing circuit. Subsequently, these signals are relayed to a computer via Wi-Fi, where a 1D convolutional neural network (1D-CNN) algorithm is used for deep learning to complete intelligent recognition[35].

By using a single-channel TSPF-Ring to slide over different contact surfaces, intelligent recognition of frictional surface features can be achieved (Fig. 3b) due to the advantages of sensors based on the triboelectric effect. Four types of textures (Fig. 3b(i)) and six types of surface materials (Fig. 3b(ii)) were selected to validate the recognition performance. The data collection process is demonstrated in Video S2. Thanks to the high sensitivity of the TSPF-Ring sensor, distinct signals can be generated when the sensor slides over the same material surface with different textures, such as smooth, textured, 2 mm-structured, and 4 mm structured patterns (Fig. 3c). These signal variations enable a high recognition accuracy of 99.28 % (Fig. 3d), with the DL architecture detailed in Table S1. When the surface materials are different, the sensor produces distinct signal amplitudes and waveforms during the sliding process due to the variation in triboelectric properties and surface morphology (Fig. 3e). By leveraging deep learning algorithms, the signal features from different materials, which may be subtle or implicit, can be extracted to achieve a recognition accuracy of 99.38 % (Fig. 3f). The real-time recognition demonstration is shown in Video S3. Additionally, in non-contact sensing applications employing the TSPF-Ring, the signal distinctions produced as different objects approach the sensor can be harnessed for non-contact material identification. Supplementary Figure S11 shows the differences in proximity signals and the confusion matrix for recognizing eight materials, with the data collection process shown in Video S4.

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Furthermore, by arranging four TSPF-Rings in an array (Supplementary Figure S12), the distance between each sensor node and the surface of different objects changes as objects approach. This enables the intelligent recognition of varying surface shapes through non-contact proximity sensing (Fig. 3g). Initially, a block was slid across the sensor array to demonstrate how the four TSPF-Rings captured the proximity signal characteristics of the approaching object. Displayed in Supplementary Figure S13, the signals generated by the sensor array fluctuated as the block moved across, validating the effectiveness of this method in capturing the positional features of the object. Subsequently, four object surface shapes—plane, cone, ellipse, cylinder—were selected as the dataset for recognition. Fig. 3h illustrates the signal differences generated by the sensor array when the proximity surface was cylinder versus plane, with the waveforms for other shapes presented in Supplementary Figure S14. The data collection platform and experimental process are shown in Video S5. It is evident that different object surface shapes produced distinct signal variations in this non-contact sensor array. Leveraging the collected signal database, a four-channel 1D-CNN deep learning algorithm was applied to accurately identify the approaching surface shapes, achieving 100 % recognition accuracy in the test set (Fig. 3i). The real-time demonstration of this process is shown in Video S6. This result underscores that the TSPF-Ring array not only ensures high sensitivity but also enhances recognition accuracy and stability through multi-point data acquisition.

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2.4. Combined visual-tactile stimulation for sensory rehabilitation

Multisensory stimulation can effectively activate the sensorimotor cortex in patients with sensory impairments, thereby accelerating the rehabilitation process [8,36]. To verify the effect of visual-tactile synchronized stimulation on brain activation, the TSPF-Ring was employed to deliver various types of sensory stimuli to participants, including tactile-only, visual-only, and visual-tactile synchronized combined stimuli, and EEG was used to record cortical activity, simultaneously (Fig. 4a). Initially, baseline EEG signals were recorded under resting condition, after which EEG signals were categorized into three groups according to the type of stimulation: Group G1 investigated the effect of tactile stimulation via the TSPF-Ring's pneumatic feedback on EEG signals; Group G2 examined the effect of visual stimulation through the TSPF-Ring's proximity-pressure mapping on the participants' sensorimotor cortex; Group G3 studied the synergistic effect of combined visual and tactile stimulation on EEG signals. Detailed experimental procedures can be found in the methods section and Supplementary Table S2. Additionally, Fig. 4b illustrates the functional regions of the human brain[37,38].

The experimental setup, as shown in Fig. 4c, included a 64-channel EEG system from ANT, along with a WaveGuard EEG cap for recording cortical activity. Tactile stimulation was administered by inflating the air chambers of the TSPF-Ring using a pneumatic pump, while visual stimulation was achieved through a proximity-pressure distribution map displayed on a computer screen, providing visual feedback to the participants. Fig. 4d displays a close-up of the interactive interface, where varying shades of blue and red represent the proximity and pressure levels, respectively. The color of each point on the hand diagram changes according to the sensor signals, allowing patients to perceive the extent of their movement and the applied pressure through visual feedback. The response time of different components in the system are shown in Table S3. Video S7 demonstrates how the TSPF-Ring's proximity and pressure sensing is used to visualize tactile information for visual feedback during a cup-grasping task.

Two healthy participants were recruited for the validation experiment, with event-related desynchronization/synchronization (ERD/ERS) used to quantify the degree of brain activation under specific tasks or stimuli [39–41]. Video S8 presents the experimental equipment and scenes used. Detailed experimental procedures and signal processing methods are described in the methods section. Alpha waves (8–13 Hz) and Beta waves (13–30 Hz), linked to sensorimotor functions[41–44], were selected as the EEG analysis bands, with the resultant ERD/ERS brain topographic maps depicted in Fig. 4e.

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In Group G1, tactile stimulation via the TSPF-Ring's pneumatic feedback induced ERD phenomena in the Alpha and Beta bands in the sensorimotor area of both participants' brains (blue regions), indicating effective activation of the sensorimotor cortex, in line with previous finding[42,43,45]. For Group G2, visual stimulation using the proximity-pressure mapping led to a pronounced ERD in the occipital region in Participant 1's Beta band, which is primarily associated with visual processing. Meanwhile, Participant 2 exhibited partial ERD in the sensorimotor cortex. Despite differences in the specific regions of activation between participants, the results demonstrate that visual stimulation enhanced cortical activation. Within Group G3, visual-tactile combined stimulation further enhanced activation in the sensorimotor region. Compared to no stimulation or single-modal stimulation,

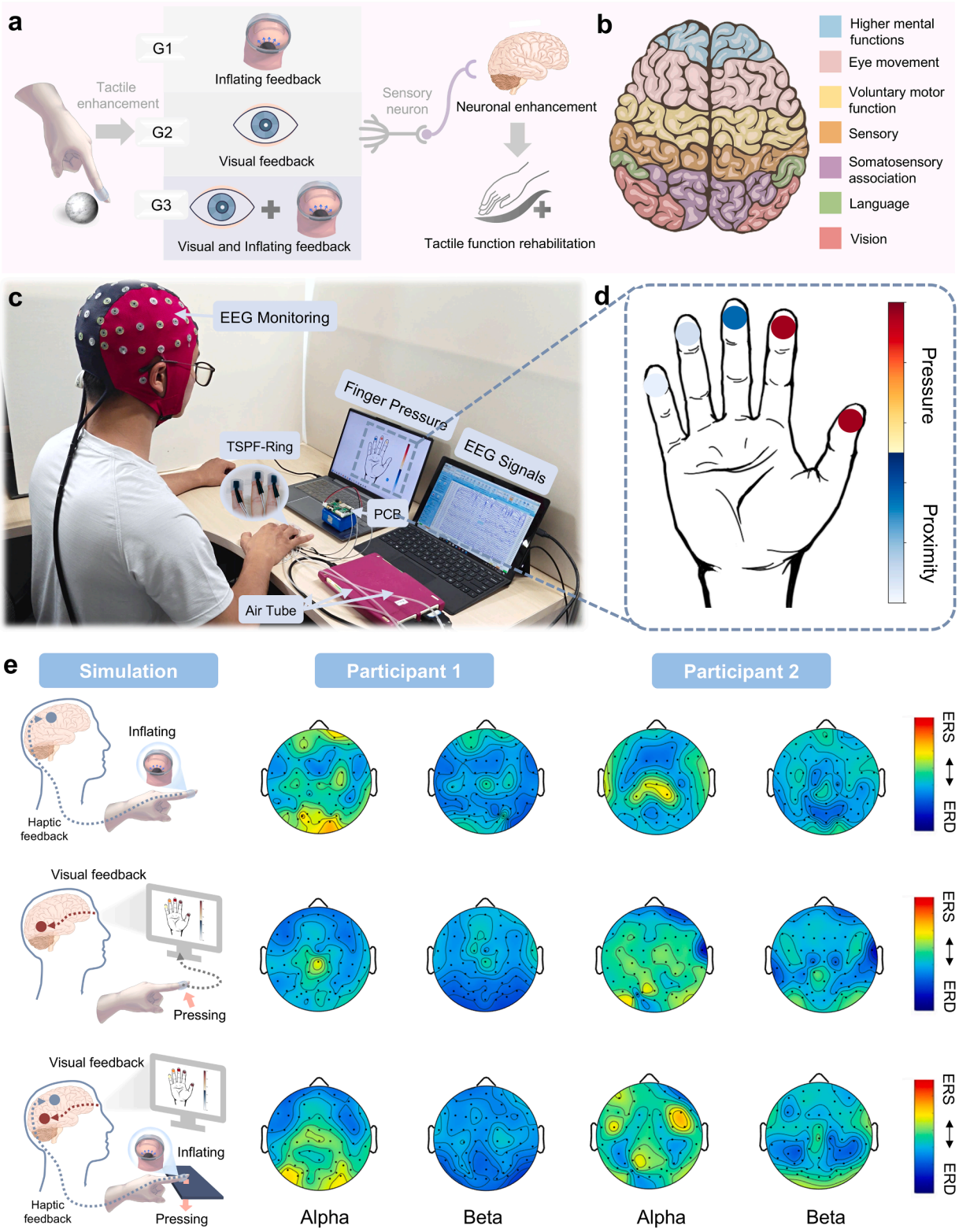


Fig. 4. Design and results of the EEG experiment. (a) Schematic diagram of the experimental protocol. (b) Functional areas of the brain. (c) Experimental setup and equipment used. (d) Enlarged view of the visualization interface. (e) Event-related desynchronization/synchronization (ERD/ ERS) brain topographic maps of alpha and beta waves in the experimental group, which indicated the power changes compared to the baseline. Blue color indicates maximal ERD, associated with brain area activation.

participants exhibited more pronounced ERD and broader activation areas in the sensorimotor cortex when exposed to combined multisensory stimulation.

These findings offer initial experimental proof that the visual-tactile combined stimulation based on the TSPF-Ring can effectively induce cortical activation in the sensorimotor region. This lays the groundwork for the development of multisensory stimulation glove systems aimed at enhancing rehabilitation for individuals with sensory impairments.

3. Conclusion

This study presents a novel TSPF-Ring system, utilizing triboelectric effect for upper limb sensory rehabilitation in stroke patients for the first time. Through the integration of multidimensional tactile sensing with pneumatic feedback, the TSPF-Ring provides an innovative approach of synchronized visual-tactile stimulation to promote neural pathway remodeling and functional recovery. Experimental validation underscores the exceptional performance of the integrated ring in proximity, pressure, and sliding detection, showcasing high sensitivity in capturing subtle tactile changes. The accuracy of tactile signal recognition surpasses 99 %, while the ring provides stable pneumatic pressure feedback within a 0–12 N adjustable range. Furthermore, EEG experiments confirm the potential of multisensory stimulation in enhancing activation in the sensorimotor cortex, providing preliminary experimental evidence for the TSPF-Ring's application in rehabilitation.

Despite the promising results achieved, further enhancement of the device and system reliability and stability is imperative to bolster its clinical utility. Subsequent efforts will focus on expanding the scope of clinical trials to assess the safety and effectiveness of visual-tactile synchronized stimulation in patient demographics. Building on this, larger-scale clinical experiments will statistically validate the efficacy of the TSPF-Ring in improving sensory and motor functions in stroke patients, ultimately facilitating its integration into practical rehabilitation therapy.

4. Experimental section

4.1. Fabrication of flexible electrode

Using a precision electronic balance, Dragonskin 30 A and B were weighed and mixed in a 1:1 mass ratio. Subsequently, MWCNT powder was added at a 100:1 mass ratio to the silicone mixture, followed by the addition of PVP dispersant at a 10:1 ratio. The resulting mixture was stirred at 1200 rpm for 3 min to ensure thorough dispersion. Finally, the prepared MWCNT/silicone mixture was uniformly applied to a 3D-printed mold, cured at room temperature for 4 h, and then demolded to obtain a flexible silicone electrode.

4.2. Fabrication of TSPF-Ring

Using a precision electronic balance, Dragonskin 30 A and B were weighed in a 1:1 mass ratio and stirred for 1 minute at 630 r/min. The mixture was then degassed under vacuum for 8 min to eliminate bubbles that could affect the sensor's performance. The degassed silicone was poured into a 3D-printed mold, and the flexible electrode was inserted, followed by the removal of excess silicone. After curing at room temperature for 4 h, the upper and lower layers of the TSPF-Ring were demolded. These layers were bonded using Sil-Poxy silicone adhesive and placed in an incubator at 60 °C for 6 h with added pressure, resulting in the completed TSPF-Ring.

4.3. Characterization and measurement

The measurements of output current and output voltage in the triboelectric sensor were performed using the Keithley 6514 system electrometer (version Keithley 6514, The Keithley Inc., USA). The signal monitoring system software platform was developed using LabVIEW and data was acquired using a data acquisition card (version NI USB-6009, Inc., USA).

4.4. Construction of the data acquisition system

First, the voltage signals from the TSPF-Ring are amplified using a precision operational amplifier in the signal processing circuit. The amplified analog voltage signals are then converted into digital signals by two ADS131M04 analog-to-digital converters (ADC) and transmitted to the main controller chip, ESP32. The ESP32 packages the data into strings and wirelessly transmits them to a computer via Wi-Fi. A TCP server on the computer receives the data from the ESP32, and a Python-based multithreaded server program handles data reception, processing, visualization, and storage, ensuring real-time accuracy and reliability of the data.

4.5. Data collection and dataset configuration

(1) Material and texture recognition: A linear motor controlled the TSPF-Ring for sliding over surfaces at 1 Hz, with signals captured at 500 Hz by a Keithley 6514. We gathered 2000 samples for material and 1200 for texture recognition, evenly distributed across categories, and split them 7:3 for training and testing. (2) Shape recognition: Objects with different shapes were moved across an array of TSPF-Rings at 1 Hz, with four-channel sensing data collected at 100 Hz via a custom board. We collected 540 samples evenly across 4 shapes, also split 7:3 for training and testing. All waveforms were subjected to a 45 Hz low-pass filter before further use to eliminate the interference of high-frequency noise.

4.6. EEG experimental setup

Group G1: Participants received tactile feedback on the fingertips via the pneumatic feedback function of the TSPF-Ring.

Group G2: The TSPF-Ring was applied with slight pressure, and participants observed a real-time pressure distribution map of their fingertips on a computer screen, serving as visual stimulation.

Group G3: Participants were instructed to actively control their fingers at a frequency of 1 Hz, gently tapping their fingertips on the table. During this task, they received pneumatic feedback from the TSPF-Ring while simultaneously observing the pressure distribution on the computer screen. This task combined active motor activity with multisensory stimulation to explore the interaction between visual-tactile synchronized stimulation and cortical activation during active movement.

4.7. EEG data acquisition

Calibration was performed by recording 30 seconds of EEG data while participants were at rest, ensuring sufficient samples to capture baseline EEG characteristics. During the experimental tasks, 10 seconds of EEG data were recorded during each stimulation session, with a 30-second rest period between tasks, for a total of three cycles. This design ensured the collection of adequate EEG data during stimulation while allowing participants sufficient rest between tasks to prevent fatigue and adaptation effects that might influence results.

4.8. EEG data processing

EEG data were processed and analyzed using the open-source MATLAB toolbox, EEGLAB.

4.8.1. Preprocessing steps

(1) Remove 50 Hz power line noise. (2) Apply a high-pass filter to remove baseline drift below 1 Hz. (3) Apply a band-pass filter to retain signals between 1 and 45 Hz. (4) Re-reference the data using average reference. (5) Use Independent Component Analysis (ICA) to remove eye blink artifacts. (6) Extract 10 seconds of data for pre-stimulation, during stimulation, and post-stimulation periods.

4.8.2. Power spectral density (PSD) analysis

(1) Each 10-second segment was divided into 1-second epochs. PSD was calculated for each epoch using the Welch method with a 50 % overlap window and Hanning window applied. (2) The PSD results from

all 1-second epochs were averaged to obtain the mean PSD for the entire 10-second period.

4.8.3. EEG frequency band analysis

This study focused on the Alpha band (8–13 Hz) and Beta band (13–30 Hz), particularly the event-related desynchronization (ERD) phenomena in sensorimotor tasks. Event-related synchronization (ERS) typically indicates cortical inhibition, while ERD is associated with cortical activation. The oscillatory patterns in the Alpha and Beta bands reflect the characteristic spatiotemporal dynamics of the brain during sensorimotor tasks, providing effective indicators for evaluating the impact of sensorimotor training on cortical activation. The ERD calculation is expressed as:

$$ERD\ ratio_{i,f} = \frac{PSD_{i,f,t} - PSD_{i,f,b}}{PSD_{i,f,b}}$$

where i represents the EEG channel, f represents the frequency band, $PSD_{i,f,t}$ is the mean PSD during stimulation, and $PSD_{i,f,b}$ is the mean PSD during the resting baseline period.

CRediT authorship contribution statement

Bo Yang: Software, Methodology, Investigation, Formal analysis. **Lifa Yang**: Methodology, Investigation, Formal analysis. **Haohan Zhao**: Methodology. **Feiyu Pan**: Methodology. **Xiangrong Cheng**: Methodology. **Linhong Ji**: Supervision. **Xingyu Wang**: Formal analysis. **Chong Li**: Supervision, Formal analysis. **Wei Li**: Supervision, Conceptualization. **Xuecheng Qu**: Supervision, Funding acquisition. **Jia Cheng**: Supervision, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.nanoen.2024.110638.

Data availability

Data will be made available on request.

References

[1] N.A. Hilkens, B. Casolla, T.W. Leung, F.-E. De Leeuw, Stroke, Lancet 403 (2024) 2820–2836, https://doi.org/10.1016/S0140-6736(24)00642-1.
[2] S. Wu, B. Wu, M. Liu, Z. Chen, W. Wang, C.S. Anderson, P. Sandercock, Y. Wang, Y. Huang, L. Cui, C. Pu, J. Jia, T. Zhang, X. Liu, S. Zhang, P. Xie, D. Fan, X. Ji, K.-S. L. Wong, L. Wang, Stroke in China: advances and challenges in epidemiology, prevention, and management, Lancet Neurol. 18 (2019) 394–405, https://doi.org/10.1016/S1474-4422(18)30500-3.
[3] P. Langhorne, J. Bernhardt, G. Kwakkel, Stroke rehabilitation, Lancet 377 (2011) 1693–1702, https://doi.org/10.1016/S0140-6736(11)60325-5.
[4] Q. Xie, W. Lin, Q. Zhang, B. Sheng, Y. Zhang, J. Huang, Research progress of intelligent evaluation and virtual reality based training in upper limb rehabilitation afrer stroke, Rehabil. Med. 33 (2023) 271–279, https://doi.org/10.3724/SP.J.1329.2023.03011.
[5] S.S. Kessner, E. Schlemm, B. Cheng, U. Bingel, J. Fiehler, C. Gerloff, G. Thomalla, Somatosensory deficits after ischemic stroke: time course and association with infarct location, Stroke 50 (2019) 1116–1123, https://doi.org/10.1161/STROKEAHA.118.023750.
[6] S.B. Zandvliet, G. Kwakkel, R.H.M. Nijland, E.E.H. Van Wegen, C.G.M. Meskers, Is recovery of somatosensory impairment conditional for upper-limb motor recovery early after stroke? Neurorehabil. Neural Repair 34 (2020) 403–416, https://doi.org/10.1177/1545968320907075.
[7] N. Bolognini, C. Russo, D.J. Edwards, The sensory side of post-stroke motor rehabilitation, Restor. Neurol. Neurosci. 34 (2016) 571–586, https://doi.org/10.3233/RNN-150606.
[8] Y.-H. Hung, H.-Y. Lai, C.-H. Shih, Study of multi-sensory stimulation for the design of hand rehabilitation equipment for stroke patients, J. Ind. Prod. Eng. 32 (2015) 425–431, https://doi.org/10.1080/21681015.2015.1058296.
[9] S. Zhang, Y. Yu, P. Xu, X. Shen, C. Fang, X. Wu, P. Qu, T. Wu, Q.M. Wang, X. Luo, Y. Hong, Mechanical digit sensory stimulation: a randomized control trial on neurological and motor recovery in acute stroke, Front. Neurosci. 17 (2023) 1134904, https://doi.org/10.3389/fnins.2023.1134904.
[10] J. Wang, B. Wu, Y. Tong, X. Wang, Z. Lu, W. Wang, Effect of acupuncture combined with rehabilitation training on sensory impairment of patients with stroke: a network meta-analysis, BMC Complement. Med. Ther. 24 (2024) 102, https://doi.org/10.1186/s12906-024-04401-9.
[11] L. Lu, G. Hu, J. Liu, B. Yang, 5G NB-IoT system integrated with high-performance fiber sensor inspired by cirrus and spider structures, Adv. Sci. 11 (2024) 2309894, https://doi.org/10.1002/advs.202309894.
[12] J. Li, C. Carlos, H. Zhou, J. Sui, Y. Wang, Z. Silva-Pedraza, F. Yang, Y. Dong, Z. Zhang, T.A. Hacker, B. Liu, Y. Mao, X. Wang, Stretchable piezoelectric biocrystal thin films, Nat. Commun. 14 (2023) 6562, https://doi.org/10.1038/s41467-023-42184-8.
[13] T. Feng, D. Ling, C. Li, W. Zheng, S. Zhang, C. Li, A. Emel'yanov, A.S. Pozdnyakov, L. Lu, Y. Mao, Stretchable on-skin touchless screen sensor enabled by ionic hydrogel, Nano Res. 17 (2024) 4462–4470, https://doi.org/10.1007/s12274-023-6365-8.
[14] N. Van Damme, R. Ratz, L. Marchal-Crespo, Towards unsupervised rehabilitation: development of a portable compliant device for sensorimotor hand rehabilitation, in: Proceedings of the International Conference on Rehabil. Robot., ICORR, IEEE, Rotterdam, Netherlands, 2022, pp. 1–6, https://doi.org/10.1109/ICORR55369.2022.9896556.
[15] Y. Gu, Y. Xu, Y. Shen, H. Huang, T. Liu, L. Jin, H. Ren, J. Wang, A review of hand function rehabilitation systems based on hand motion recognition devices and artificial intelligence, Brain Sci. 12 (2022) 1079, https://doi.org/10.3390/brainsci12081079.
[16] H. Rodgers, H. Bosomworth, H.I. Krebs, F. Van Wijck, D. Howel, N. Wilson, L. Aird, N. Alvarado, S. Andole, D.L. Cohen, J. Dawson, C. Fernandez-Garcia, T. Finch, G. A. Ford, R. Francis, S. Hogg, N. Hughes, C.I. Price, L. Ternent, D.L. Turner, L. Vale, S. Wilkes, L. Shaw, Robot assisted training for the upper limb after stroke (RATULS): a multicentre randomised controlled trial, Lancet 394 (2019) 51–62, https://doi.org/10.1016/S0140-6736(19)31055-4.
[17] M. Sui, Y. Ouyang, H. Jin, Z. Chai, C. Wei, J. Li, M. Xu, W. Li, L. Wang, S. Zhang, A soft-packaged and portable rehabilitation glove capable of closed-loop fine motor skills, Nat. Mach. Intell. 5 (2023) 1149–1160, https://doi.org/10.1038/s42256-023-00728-z.
[18] Y. Murakami, K. Honaga, H. Kono, K. Haruyama, T. Yamaguchi, M. Tani, R. Isayama, T. Takakura, A. Tanuma, K. Hatori, F. Wada, T. Fujiwara, New artificial intelligence-integrated electromyography-driven robot hand for upper extremity rehabilitation of patients with stroke: a randomized, controlled trial, Neurorehabil. Neural Repair 37 (2023) 298–306, https://doi.org/10.1177/15459683231166939.
[19] C. Nam, W. Rong, W. Li, C. Cheung, W. Ngai, T. Cheung, M. Pang, L. Li, J. Hu, H. Wai, X. Hu, An exoneuromusculoskeleton for self-help upper limb rehabilitation after stroke, Soft Robot 9 (2022) 14–35, https://doi.org/10.1089/soro.2020.0090.
[20] E.R. Goldenkoff, H.R. McGregor, J. Mergos, P. Gholizadeh, J. Bridenstine, M.J. N. Brown, M. Vesia, Reversal of visual feedback modulates somatosensory plasticity, Neuroscience 452 (2021) 335–344, https://doi.org/10.1016/j.neuroscience.2020.10.033.
[21] M.V. Pham, K. Saito, S. Miyaguchi, H. Watanabe, H. Ikarashi, K. Nagasaka, H. Yokota, S. Kojima, Y. Inukai, N. Otsuru, H. Onishi, Changes in excitability and GABAergic neuronal activity of the primary somatosensory cortex after motor learning, Front. Neurosci. 16 (2022) 794173, https://doi.org/10.3389/fnins.2022.794173.
[22] X. Qu, Z. Liu, P. Tan, C. Wang, Y. Liu, H. Feng, D. Luo, Z. Li, Z.L. Wang, Artificial tactile perception smart finger for material identification based on triboelectric sensing, Sci. Adv. 8 (2022) eabq2521, https://doi.org/10.1126/sciadv.abq2521.
[23] X. Qu, J. Xue, Y. Liu, W. Rao, Z. Liu, Z. Li, Fingerprint-shaped triboelectric tactile sensor, Nano Energy 98 (2022) 107324, https://doi.org/10.1016/j.nanoen.2022.107324.
[24] Z. Sun, M. Zhu, X. Shan, C. Lee, Augmented tactile-perception and haptic-feedback rings as human-machine interfaces aiming for immersive interactions, Nat. Commun. 13 (2022) 5224, https://doi.org/10.1038/s41467-022-32745-8.
[25] S. Lin, W. Yang, X. Zhu, Y. Lan, K. Li, Q. Zhang, Y. Li, C. Hou, H. Wang, Triboelectric micro-flexure-sensitive fiber electronics, Nat. Commun. 15 (2024) 2374, https://doi.org/10.1038/s41467-024-46516-0.
[26] X. Qu, S. Cheng, Y. Liu, Y. Hu, Y. Shan, R. Luo, S. Weng, H. Li, H. Niu, M. Gu, Y. Fan, B. Shi, Z. Liu, W. Hua, Z. Li, Z.L. Wang, Bias-free cardiac monitoring

capsule, Adv. Mater. 36 (2024) 2402457, https://doi.org/10.1002/adma.202402457.
[27] Y. Liu, C. Yiu, Z. Song, Y. Huang, K. Yao, T. Wong, J. Zhou, L. Zhao, X. Huang, S. K. Nejad, M. Wu, D. Li, J. He, X. Guo, J. Yu, X. Feng, Z. Xie, X. Yu, Electronic skin as wireless human-machine interfaces for robotic VR, Sci. Adv. 8 (2022) eabl6700, https://doi.org/10.1126/sciadv.abl6700.
[28] Y. Luo, C. Liu, Y.J. Lee, J. DelPreto, K. Wu, M. Foshey, D. Rus, T. Palacios, Y. Li, A. Torralba, W. Matusik, Adaptive tactile interaction transfer via digitally embroidered smart gloves, Nat Commun 15 (2024) 868, https://doi.org/10.1038/s41467-024-45059-8.
[29] E. Vendrame, A. Coletti, L. Cappello, E. Mastinu, C. Cipriani, A Wearable Device for Hand Sensorimotor Rehabilitation Through Augmented Sensory Feedback, in: Proceedings of the International Conference on Rehabil. Robot., ICORR, IEEE, Singapore, Singapore, 2023, pp. 1–6, https://doi.org/10.1109/ICORR58425.2023.10304722.
[30] X. Chen, L. Gong, L. Wei, S.-C. Yeh, L. Da Xu, L. Zheng, Z. Zou, A wearable hand rehabilitation system with soft gloves, IEEE Trans. Ind. Inform. 17 (2021) 943–952, https://doi.org/10.1109/TII.2020.3010369.
[31] K. Kitai, M. Odagiri, R. Yamauchi, T. Kodama, Evaluation of intervention effectiveness of sensory compensatory training with tactile discrimination feedback on sensorimotor dysfunction of the hand after stroke, Brain Sci. 11 (2021) 1314, https://doi.org/10.3390/brainsci11101314.
[32] H. Liu, D. Li, H. Chu, Y. Ding, Z. Fu, X. Yao, J. Zhu, J. Yang, R. Liu, T. Xu, S. Fu, Y. Liu, Y. Han, Y. Wang, Y. Zhao, X. Cui, Y. Tian, Ultra-stretchable triboelectric touch pad with sandpaper micro-surfaces for Transformer-assisted gesture recognition, Nano Energy 130 (2024) 110110, https://doi.org/10.1016/j.nanoen.2024.110110.
[33] M. Zhu, Z. Sun, C. Lee, Soft modular glove with multimodal sensing and augmented haptic feedback enabled by materials' multifunctionalities, ACS NANO 16 (2022) 14097–14110, https://doi.org/10.1021/acsnano.2c04043.
[34] N.-W. Pak, J.-H. Lee, Effects of visual feedback training and visual targets on muscle activation, balancing, and walking ability in adults after hemiplegic stroke: a preliminary, randomized, controlled study, Int. J. Rehabil. Res. 43 (2020) 76–81, https://doi.org/10.1097/MRR.0000000000000376.
[35] B. Yang, J. Cheng, X. Qu, Y. Song, L. Yang, J. Shen, Z. Bai, L. Ji, Triboelectric-inertial sensing glove enhanced by charge-retained strategy for human-machine interaction, Adv. Sci. (2024) 2408689, https://doi.org/10.1002/advs.202408689.
[36] Y. Saito, K. Ohnishi, T. Komeda, T. Seki, The study on training equipment of hand and arm by mechanical and visual stimulation feedback, in: Proceedings of the 10th Fr.-Jpn. 8th Eur.-Asia Congr. Mecatronics MECATRONICS2014- Tokyo, IEEE, Tokyo, Japan, 2014, pp. 173–178, https://doi.org/10.1109/MECATRONICS.2014.7018576.
[37] S.H. Siddiqi, K.P. Kording, J. Parvizi, M.D. Fox, Causal mapping of human brain function, Nat. Rev. Neurosci. 23 (2022) 361–375, https://doi.org/10.1038/s41583-022-00583-8.
[38] D.V. Demeter, D.J. Greene, The promise of precision functional mapping for neuroimaging in psychiatry, Neuropsychopharmacology (2024), https://doi.org/10.1038/s41386-024-01941-z.
[39] G. Pfurtscheller, F.H. Lopes Da Silva, Event-related EEG/MEG synchronization and desynchronization: basic principles, Clin. Neurophysiol. 110 (1999) 1842–1857, https://doi.org/10.1016/S1388-2457(99)00141-8.
[40] M. Takahashi, K. Takeda, Y. Otaka, R. Osu, T. Hanakawa, M. Gouko, K. Ito, Event related desynchronization-modulated functional electrical stimulation system for stroke rehabilitation: a feasibility study, J. Neuroeng. Rehabil. 9 (2012) 56, https://doi.org/10.1186/1743-0003-9-56.
[41] T. Jia, J. Sun, C. McGeady, L. Ji, C. Li, Enhancing brain–computer interface performance by incorporating brain-to-brain coupling, Cyborg Bionic Syst. 5 (2024) 0116, https://doi.org/10.34133/cbsystems.0116.
[42] A. Kuc, I. Skorokhodov, A. Semirechenko, G. Khayrullina, V. Maksimenko, A. Varlamov, S. Gordleeva, A. Hramov, Oscillatory responses to tactile stimuli of different intensity, Sensors 23 (2023) 9286, https://doi.org/10.3390/s23229286.
[43] Y. Geng, L. Qin, Y. Li, Z. Yu, L. Li, M.G. Asogbon, Y. Zhan, N. Yan, X. Guo, G. Li, Identifying oscillations under multi-site sensory stimulation for high-level peripheral nerve injured patients: a pilot study, J. Neural Eng. 19 (2022) 036028, https://doi.org/10.1088/1741-2552/ac7079.
[44] R. Wilson, K.J. Mullinger, S.T. Francis, S.D. Mayhew, The relationship between negative BOLD responses and ERS and ERD of alpha/beta oscillations in visual and motor cortex, NeuroImage 199 (2019) 635–650, https://doi.org/10.1016/j.neuroimage.2019.06.009.
[45] X. Shu, S. Chen, J. Meng, L. Yao, X. Sheng, J. Jia, D. Farina, X. Zhu, Tactile stimulation improves sensorimotor rhythm-based BCI performance in stroke patients, IEEE Trans. Biomed. Eng. 66 (2019) 1987–1995, https://doi.org/10.1109/TBME.2018.2882075.